

SARANPAUL', Russian Federation

WMO: 235270

Lat: **64.253N**Lon: **60.931E**Elev: **29**StdP: **100.98**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.0	-39.2	-44.2	0.0	-41.9	-41.3	0.1	-39.1	8.1	-14.9	7.0	-7.6	0.3	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	27.5	18.7	25.5	17.5	23.5	16.5	19.8	24.8	18.8	23.7	17.6	22.1	2.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	21.8	16.9	12.1	20.7	15.6	11.1	19.8	56.8	24.8	53.3	23.7	49.6	22.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.5	5.5	-43.3	30.0	1.8	1.5	-44.5	31.0	-45.6	31.9	-46.6	32.7	-47.8	33.8		
(5)			-43.2	21.3	1.7	1.4	-44.4	22.3	-45.4	23.1	-46.4	23.9	-47.6	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.9	-24.2	-18.7	-10.7	-2.9	5.5	13.0	16.2	12.7	6.4	-1.8	-12.8	-19.2
	DBStd	15.05	9.13	8.78	8.01	5.47	5.18	4.74	4.01	3.70	3.82	5.45	8.92	10.55
	HDD10.0	5165	1059	804	641	387	161	22	3	14	120	366	683	906
	HDD18.3	7805	1317	1037	899	637	398	170	89	178	358	624	933	1164
	CDD10.0	439	0	0	0	0	21	112	197	98	12	0	0	0
	CDD18.3	38	0	0	0	0	0	11	24	4	0	0	0	0
	CDH23.3	436	0	0	0	0	17	110	273	36	1	0	0	0
	CDH26.7	81	0	0	0	0	1	19	57	3	0	0	0	0
Wind	WSAvg	2.1	1.5	1.8	2.4	2.6	2.7	2.4	2.0	2.0	2.0	1.9	1.8	1.6
Precipitation	PrecAvg	498	25	20	23	34	39	52	73	64	57	45	35	29
	PrecMax	690	59	41	61	92	90	133	202	150	151	89	90	87
	PrecMin	356	10	2	0	4	3	7	19	13	18	3	8	3
	PrecStd	89	13	9	16	18	20	35	44	33	33	21	17	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	2.2	6.4	10.6	25.7	29.3	30.6	26.5	20.7	11.2	4.2	2.1
		MCWB	-2.2	0.2	1.8	4.2	15.3	18.6	19.5	19.0	14.4	7.8	2.6	0.0
	2%	DB	-4.3	-1.0	4.1	7.8	22.1	26.2	28.7	24.1	17.2	8.7	2.5	0.4
		MCWB	-5.7	-2.3	1.0	3.0	13.0	18.0	19.3	17.3	12.2	6.1	1.0	-0.8
	5%	DB	-7.0	-3.9	2.6	6.3	19.0	24.0	26.7	21.9	14.8	6.6	1.0	-1.7
		MCWB	-8.0	-5.1	0.0	2.6	11.7	16.6	18.3	16.3	11.0	4.7	-0.6	-2.9
	10%	DB	-10.3	-5.6	0.9	5.0	15.2	21.4	24.6	19.9	12.8	4.9	-0.7	-4.0
		MCWB	-11.0	-6.7	-0.9	2.0	9.2	15.1	17.2	15.2	9.7	3.6	-2.0	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.2	0.5	2.6	5.2	16.5	20.4	21.7	20.2	14.8	8.4	2.6	0.3
		MCDB	-1.0	2.1	5.9	8.7	24.9	26.2	27.9	24.7	20.0	10.6	3.8	1.6
	2%	WB	-5.6	-2.4	1.2	4.0	14.1	19.0	20.4	18.7	12.9	6.7	1.1	-0.8
		MCDB	-4.5	-1.1	3.5	6.7	20.4	24.5	25.6	22.3	15.7	8.5	2.5	0.4
	5%	WB	-7.9	-4.9	0.3	3.1	12.2	17.5	19.2	17.1	11.5	4.9	-0.5	-2.8
		MCDB	-7.0	-3.9	2.4	5.9	18.2	22.4	24.2	20.5	14.4	6.1	0.8	-1.9
	10%	WB	-11.0	-6.7	-1.0	2.1	9.9	15.8	18.3	15.9	10.1	3.6	-2.1	-5.0
		MCDB	-10.3	-5.6	1.0	4.8	15.1	20.3	23.3	19.3	12.4	4.8	-0.9	-4.1
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	11.4	9.9	10.3	10.0	11.1	10.3	8.0	6.7	7.9	8.4
		MCDBR	10.1	9.4	9.3	10.4	16.7	13.4	15.0	13.6	12.1	8.5	7.9	7.6
		MCWBR	9.5	8.4	7.0	6.9	9.1	6.1	6.4	7.3	7.6	5.9	6.9	6.9
	5% WB	MCDBR	10.1	9.6	8.7	8.9	15.0	11.8	12.7	11.2	10.7	6.9	7.5	7.6
		MCWBR	9.5	8.7	6.9	6.1	8.9	6.1	6.0	6.8	7.1	5.0	6.8	7.1
Clear-Sky Solar Irradiance	taub	0.210	0.272	0.313	0.344	0.353	0.366	0.410	0.373	0.330	0.290	0.207	0.211	
	taud	1.924	2.069	2.082	2.127	2.352	2.438	2.311	2.395	2.482	2.359	1.968	1.919	
	Ebn at Noon	333	618	752	819	856	852	793	793	760	623	330	111	
	Edh at Noon	28	64	98	119	105	99	110	92	69	49	25	11	
All-Sky Solar Radiation	RadAvg	0.16	0.83	2.26	3.95	5.17	5.22	5.33	3.63	1.93	0.87	0.23	0.06	
	RadStd	0.01	0.07	0.19	0.36	0.43	0.42	0.42	0.35	0.24	0.10	0.03	0.00	